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WHAT COULD WE LEARN FROM OBSERVATIONS OF THE MUON CHARGE RATIO IN COSMIC RAY AIR SHOWERS ?

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Abstract. Various aspects of studies of the charge ratio of atmospheric muons are reviewed, and in particular the muon charge ratio of the lateral muon density distributions in single EAS is considered in context of recent proposals to measure this observable triggered by EAS observations.

Key words: cosmic rays, muon charge ratio, geomagnetic field effects.

1. INTRODUCTION

Primary cosmic rays are dominantly high-energy protons, alpha particles and heavier nuclei in a relative amount decreasing with the atomic number. When penetrating from the outer space into the Earth's atmosphere they initiate the development of a phenomenon called Extensive Air Showers (EAS) by multiple production of particles in cascading interactions of the primary particles with atmospheric nuclei. The produced secondary radiation establishes an essential feature of our natural environment. It affects material and biological substances and comprises a specific part of the natural radiation background. Its study is of relevance in various scientific problems. Photons, electrons and positrons are the most numerous secondary particles in an EAS event, and the muonic component contributes only to few percent in the single shower. However, at lower primary energies, which are dominating due to the steeply falling primary spectrum, or in case of very inclined showers the electromagnetic shower component gets completely absorbed during the

travel through the atmosphere, while the muons ("penetrating component") survive the propagation through even larger slant depths. Hence the inclusive secondary radiation flux in the atmosphere comprises mainly muons (ca. 80 % with ca. 100 muons $s^{-1}m^{-2}sr^{-1}$ on sea level).

The cosmic ray muons originate from decay of hadronic secondaries produced in particle cascades by primary cosmic rays:

$$\pi^{\pm} \rightarrow \mu^{\pm} + \nu_{\mu}(\overline{\nu}_{\mu}) \text{ 100.0 \% (lifetime } 2.6 \cdot 10^{-2}\mu\text{s)}$$
$$K^{\pm} \rightarrow \mu^{\pm} + \nu_{\mu}(\overline{\nu}_{\mu}) \text{ 63.5 \% (lifetime } 1.2 \cdot 10^{-2}\mu\text{s).}$$

The muons decay with a larger life time ($2.2\,\mu$s):

$$\mu^{+} \rightarrow e^{+} + \nu_{e} + \overline{\nu}_{\mu}, \quad \mu^{-} \rightarrow e^{-} + \overline{\nu}_{e} + \nu_{\mu}.$$

The ratio of positive to negative muons, the so-called muon charge ratio $R_{\mu}(\mu^{+}/\mu^{-})$ is a significant quantity which reflects important features of the hadronic meson production in cosmic ray collisions [1, 2] and can help to discern the primary mass composition [2, 3]. It is also immediately obvious that the muon flux in the atmosphere is strongly related to the neutrino flux and that the muon charge ratio

$$R_{\mu}(\mu^{+}/\mu^{-}) \sim R(\nu_{e}/\overline{\nu}_{e})$$

provides relevant information for neutrino anomaly. The atmospheric neutrino anomaly is the observation of Super-Kamiokande [4] and other experiments [5] that the ratio of muonic to electronic neutrinos is much smaller than the theoretical predictions, $R(\nu_{\mu}/\nu_{e})_{observed}/R(\nu_{\mu}/\nu_{e})_{predicted} \ll 1$. The deficit is interpreted in terms of neutrino flavour oscillations.

Several studies have re-examined the influence of the Earth's magnetic field on muon and neutrino fluxes [6, 7]. At lower energies, the fluxes of the primaries and secondaries are influenced by the Earth's magnetic field, in particular by a magnetic rigidity cut-off of the primary cosmic rays penetrating the Earth's atmosphere from the cosmos [7], resulting in latitude dependence and in an east-west asymmetry of the incident flux [8]. There are numerous studies of the charge ratio of the atmospheric muon flux, on sea level (see the compilations [2, 9–11]) and also of the vertical dependence by balloon experiments e.g. [12–14]. They provide a highly inclusive information since the atmospheric flux is produced by many different EAS, with primary energies distributed along the steeply falling energy spectrum and the mass composition of the primary flux. When impinging on our Earth's atmosphere they are additionally affected by the geomagnetic field. The influence of the geomagnetic field leads to a dependence of the muon charge ratio from the azimuth of the direction of incidence, in particular for low energy muons, which are dominantly originating of EAS at lower primary energies. The measured value of muon charge ratio which is empirically found to have a value of about

1.25–1.30, is mainly a result of the positive charge (proton) excess of the primary mass distribution, at $E_\mu > 10\,\mathrm{GeV}$ slightly increasing with the energy of the observed muons [11].

In the charge ratio of muons of single EAS (exclusive observations), which could be specified by a definite primary energy, the direction of incidence and eventually by the mass of the primary, the effects of the type of the hadronic interaction and geomagnetic field are expected to get revealed in a more pronounced way. In particular the lateral distribution of the EAS particles displays an azimuthal variation [15], which is influenced by the geomagnetic field, differently for μ^+ and μ^-, and leading to an azimuthal variation of the charge ratio of the muon density distribution.

In this communication we address some aspects of the muon charge ratio, especially of the azimuthal variations of the muon density distribution, worked out by simulation studies. The considerations are in some context of recent proposals for studies of the muon charge ratio in single EAS, which will be observed with standard detector arrays as used (see [16]) to study EAS.

2. ASPECTS OF HADRONIC INTERACTIONS

Let us first to throw some light by the standard analytical approach of calculating the muon spectra and muon charge ratio of the atmospheric flux related to the hadronic interaction properties [17, 18, 19].

In the approximation of the diffusion equations of the propagation of the particles through the atmosphere the fluxes of the particles are governed by the spectrum weighted moments Z of the inclusive cross sections

$$Z_{ac} = \int (x_F)^{\gamma-1}\, F_{ac}(x_F)\, dx_F,$$

$(x_F)^{\gamma-1} = (E_c/E_a)^{\gamma-1}$ relative energy transfer spectrum,

$F_{ac}(E_c, E_a) = E_c\ dn_c(E_c, E_a)/dE_c,$

dn_c = the number of particles c produced in the energy bin dE_c,

$\gamma - 1$ = the power of the integral energy spectrum, $\gamma = 2.7$,

the indices a and c indicate the entrance and exit channels.

For the flux of pions und muons the moments $Z_{p\pi+/-}$ and $Z_{n\pi+/-}$ are of importance with the isospin symmetry

$$Z_{p\pi+} = Z_{n\pi-}, \quad Z_{p\pi-} = Z_{n\pi+}.$$

As $Z_{p\pi+} > Z_{n\pi-}$ (see [18]) the value of the charge ratio does not just reflect the positive charge excess, but also the difference in the interaction producing pions of different charges. This is a consequence of the valence quark content of protons and pions. There are further source terms for the pion flux in the atmosphere represented by

$$Z_{\pi+\pi+} = Z_{\pi-\pi-} \quad \text{and} \quad Z_{\pi+\pi-} = Z_{\pi-\pi+}.$$

The contribution of subsequent pion interactions to the muon flux is affected by the decay probability of the π-mesons. At low energies the source of kaons are proton and neutron fluxes. The asymmetry in K^+- and K^--production is enhanced by $Z_{pK+} \neq Z_{nK-}$. K^+ and K^- do not belong to the same isospin multiplet since the $K^- = [\, s\,\overline{u}\,]$ does not have any contribution from the valence quark content of the incident nucleon, reflected by $Z_{pK+} \ll Z_{nK-} \approx Z_{pK-}$. Thus it is expected that the charge ratio of muons will increase at higher energies due to the enhanced contribution of the K^+ decay.

The approach, though applicable with some assumptions and approximations, is enabled by the fact that power law dependencies of the uncorrelated fluxes enter, and it has the advantage that measured inclusive cross sections could be introduced, as far they are known. Modern theoretical approaches of the muon flux and charge ratio start from Monte Carlo simulations of a sufficient number of single EAS events calculated along the primary energy spectrum and chemical distribution of cosmic rays (see [1] e.g.). The Monte Carlo simulations invoke as generators specific models of the hadronic interactions, which are reflected by the energy-weighted moments. Therefore measurements of the charge ratio are a source of information about the validity of such hadronic interaction models. In case of the atmospheric muons it has turned out that the results are rather stable against modifications of the models, except when higher energy muons would be considered. Michael Unger (L3 Collaboration) has put attention [20] that for muons with $E_\mu > 300$ GeV the charge ratio would be very discriminative, also due to the increasing influence of kaons. Such data of sufficient precision are actually missing (see also [21]). Accurate measurements, though requiring painstaking experimental patience, would be a challenging task of a detector installed in an underground laboratory.

The features of the particular models are revealed more distinctly, when the charge ratio of single showers is studied. Fig. 1 compares the x_F-distributions of positive and negative π-mesons from proton collisions. Apparently there is an excess of positive pions already in the single collision process, as mentioned above with explaining the observation of $Z_{p\pi+} > Z_{p\pi-}$ which leads already in a single shower to a value of $R_\mu(\mu^+/\mu^-) > 1$. Some exploratory calculations for proton induced EAS of 10^{15} eV find a μ^+-excess of less then about 10 %.

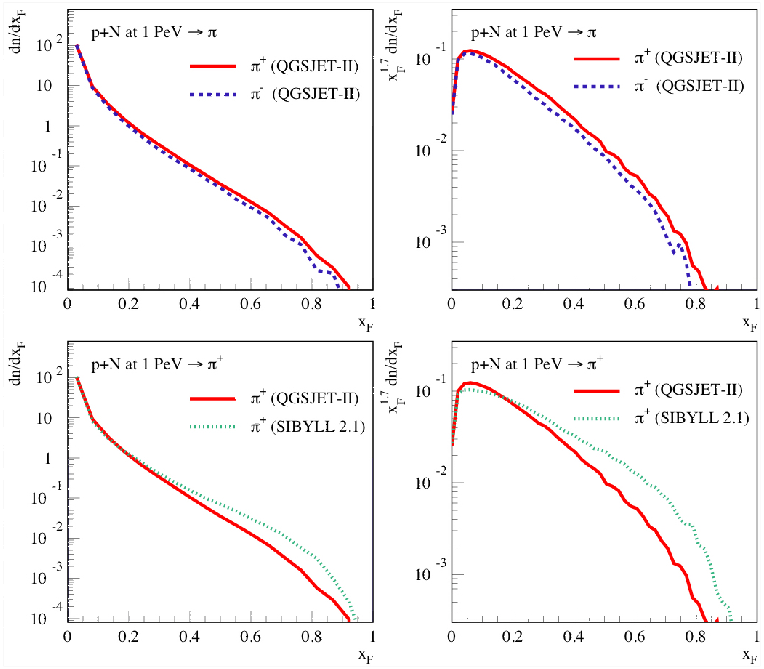


Fig. 1 – Feynman x_F distributions dn/dx_F (left) and of the integrand of the energy-weighted moments $Z_{p\pi+}$ and $Z_{p\pi-}$ (right) for the positive and negative π in proton-nitrogen-collisions at an energy of 10^{15} eV. Different hadronic interaction models [22, 23] are compared (lower panel).

Since for atmospheric muons the single distribution is weighted by $(x_F)^{-1}$ and additionally mainly the primary energy range of $10^{12} - 10^{14}$ eV is of importance, a range of higher x_F values is contributing than for a single shower of 10^{15} eV where smaller x_F get dominating. Just for comparison Fig. 2 shows the x_F distributions for the kaons indicating the difference for kaons of opposite charges. The model dependence is well pronounced when comparing the results for QGSJET [22] and SIBYLL [23]. The difference between K^+- and K^--distributions may be underestimated in the QGSJET model and overestimated in the SIBYLL model (private communication by the authors of the models).

3. GEOMAGNETIC EFFECTS OF PRIMARY COSMIC RAYS AND EAS PROPAGATION

The observation of the influence of the geomagnetic field on cosmic rays has been published in 1933 [8]. The fluxes of positively charged primary cosmic

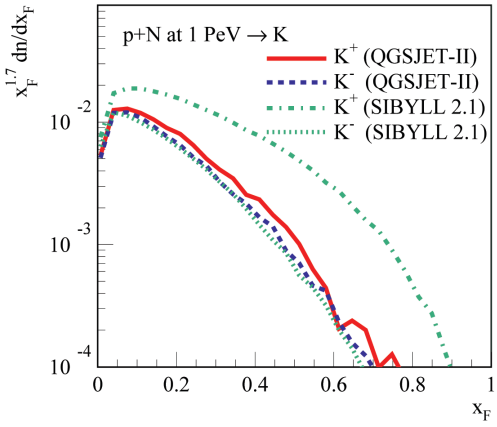


Fig. 2 – Feynman x_F distributions of the integrand of Z_{pK} for positive and negative kaons, comparing QGSJET and SIBYLL predictions.

rays that arrive the Earth's atmosphere comprise more particles travelling from west towards east than in the opposite direction and demonstrate that the dominant component of cosmic rays is positively charged. The feature is due to the magnetic field of the Earth, which does not allow that lowest rigidity particles (below a certain energy cut off) are able to reach the Earth's surface. The east-west effect, whose detailed calculation has been discussed in several studies, e.g. in [1, 7, 24, 25] is transferred to the secondary particles of air showers induced in the atmosphere, in particular to muons and to neutrinos [6, 26, 27]. Fig. 2 displays schematically, how the different bending of the trajectories of positively and negatively charged muons by the geomagnetic field contributes to the east-west asymmetry of the muon charge ratio.

The azimuthal variation of the muon charge ratio of atmospheric muons has been recently measured for low-energy inclusive muons for the latitude of the Bucharest (Romania) with a geomagnetic cut off of 5.6 GeV [28]. The experimental findings agree well with theoretical predictions calculated by use of advanced procedures [1].

As indicated in Fig. 3 apparently positively charged muons travelling towards east are bent "up" while bent "down", when travelling to west; for negative muons the situation is just opposite. Thus also in a single shower in presence of the geomagnetic field, positive and negative muons would bend in opposite directions perpendicular to the shower axis. The magnetic deviation would result in two separate centres of both muonic components.

In addition the muons of both components experience lateral dispersions arising from the transverse momenta and from multiple scattering and resulting in a distinct double-lobbed pattern of the lateral distribution in the plane

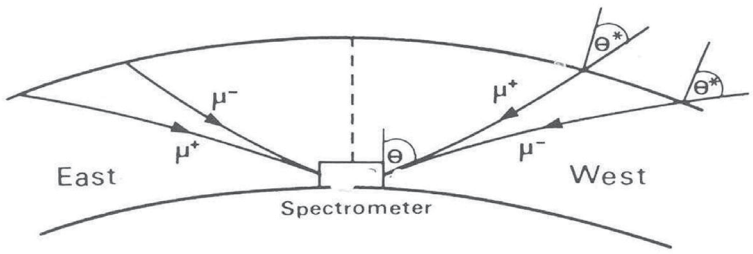


Fig. 3 – Schematic explanation of the east-west effect of the charge ratio of atmospheric muons. The positive and negative muons observed with a spectrometer e.g. under a particular zenith angle θ stem from different showers and different production directions due to the different magnetic bending of the muons of different charges. The magnetic bending of muons of different charges is different for muons travelling from east or from west, respectively. Thus different path lengths of muons change the decay probability and also the attenuation.

normal to the shower axis. The effect would depend on the direction of incidence and increases with the inclination of shower incidence i.e. when the distance of muon travel is increased due to the $\sec(\theta)$-dependence. Recently, in view of the possibility to extract charge information of high energy EAS muons the geomagnetic effects on the shower development have been estimated in [29, 30] on basis of a modified Heitler model [31], originally known as "toy" model [32]. In the following section some results on the lateral muon distributions are presented resulting from extensive 3-dimensional Monte Carlo simulations with reconstructions of the azimuthal variation of the muon component of inclined ($\theta = 45°$) showers with the primary energies of 10^{14} eV and 10^{15} eV.

4. AZIMUTHAL EFFECTS OF LATERAL DISTRIBUTIONS OF EAS MUONS

Azimuthal asymmetries and the azimuthal variation of the lateral distributions of charged EAS particles originate dominantly from the attenuation and geometrical effects of showers with inclined incidence. For a qualitative explanation we may assume that the EAS start from infinity and neglect any influence of the geomagnetic field in a first step of the argumentation. Thus we have cylindrical symmetry around the shower axis for all radial distances from the axis. For inclined showers hitting the observation plane, charged particles arriving first ("early" azimuthal EAS region) do experience less attenuation than the particles arriving later ("late" azimuthal EAS region), as the latter have travelled larger distances in atmosphere.

The azimuthal asymmetries are also present when transforming the observed particle densities to the more relevant normal shower plane, orthogonal to the axis. Recently [15] in context of investigations of the charge particle EAS component, registered by the measurements of the KASCADE-Grande experiment [33], the azimuthal asymmetries have been demonstrated for EAS of inclined incidence of the primary energy range of 10^{17} eV.

An example is displayed in Fig. 4. It shows for a particular case of the

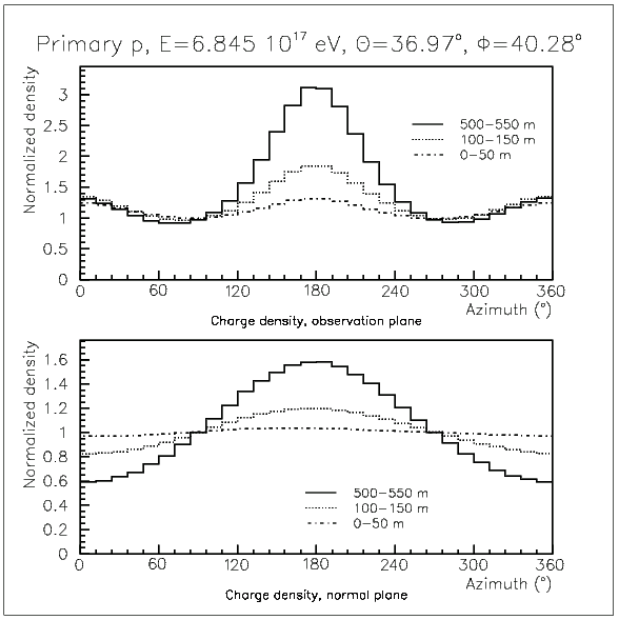


Fig. 4 – The azimuthal distributions of the normalised charged particle density of high energy proton induced EAS in the observation plane and in the plane normal to the shower axis, displaying the variation with the azimuth ϕ.

direction of EAS incidence the azimuthal variation of the charge particle density for various (radial) distances from the shower centre in the plane, normal to the shower axis and in the observation plane. For sake of convenient comparison the distributions for different radial distances have been normalized to the mean value of the charge particle densities at $\phi = 90°$ and $\phi = 270°$, where the azimuth angles (are counted from the direction of the intersection of the shower plane with the observation plane and the normal plane, respectively ($\phi = 0°$ corresponds to the "late" region of the shower).

The azimuthal dependence in the normal plane reflects mainly the effect of attenuation, while in the observation plane geometric effects are superimposed (points located in the observation plane at the same distance R from

the shower core correspond to points located in the normal plane at distances from the core from $R\cos\theta$ to R when ϕ changes from 0° to 90°).

In order to explore the azimuthal asymmetries of the lateral distributions, in particular of the muon component, some realistic and detailed Monte Carlo simulations of the EAS development have been performed. For the simulations the program CORSIKA [34] (version 6.5) has been used, invoking different models of the hadronic interaction, in particular QGSJET [22]. For particle interactions below $E_{lab} = 80$ GeV the GHEISHA option [35] is used. The earth magnetic field with a homogeneous field approximation, observation level and the particle energy thresholds have been chosen according to the conditions of the KASCADE-Grande experiment ($E_{\mu}^{thres} = 100$ MeV). The U.S. standard atmosphere (see [34]) has been adopted. For a realistic description the electron-photon component is simulated by the EGS option [36].

The performed simulations comprise proton and Fe induced EAS, with a zenith angle of incidence of $\theta = 45°$ and arriving from different cardinal points: north, east, south, west. About 1 000 showers have been simulated for most cases. For each event the muon lateral distributions for both muon charges separately and the distribution of the charge ratio of the muon density have been reconstructed. In order to explore the influence of the geomagnetic field, the simulations have been also done with the Earth's magnetic field switched off.

The following figures (Figs. 5–11) display a selection of results, mainly

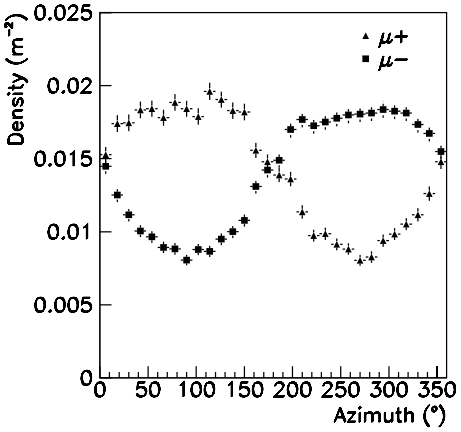


Fig. 5 – The mean azimuthal μ^+- and μ^--distributions of proton induced EAS of inclined showers ($\theta = 45°$) incident from north with the primary energy of 10^{15} eV at a distance of $45 - 50$ m from the shower axis.

for the primary energy of 10^{15} eV, since the EAS of that energy have a muon content large enough leading to a sufficient statistical accuracy. Only the case of protons is displayed as primary Fe shows similar features. The distributions

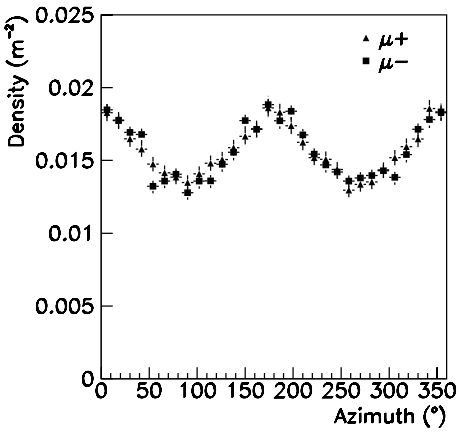


Fig. 6 – The mean azimuthal μ^+- and μ^--distributions of proton induced EAS of inclined showers ($\theta = 45°$) incident from north with the primary energy of 10^{15} eV at a distance of $45 - 50$ m from the shower axis. Earth's magnetic field switched off.

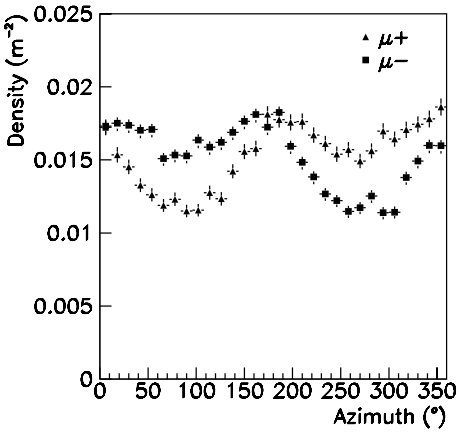


Fig. 7 – The mean azimuthal μ^+- and μ^--distributions of proton induced EAS of inclined showers ($\theta = 45°$) incident from south with the primary energy of 10^{15} eV at a distance of $45 - 50$ m from the shower axis.

are dislayed in the observation plane as a function of the azimuth angle ϕ around the shower core.

From the features shown in Figs. 5–11 can be understood when taking into account the direction of the geomagnetic vector (for the location of Karlsruhe for which the calculations are made, the magnetic inclination has a value of about 65° north i.e. the vector points downwards). From these features and from some tentative results studies at lower primary energies we deduce the following conclusions:

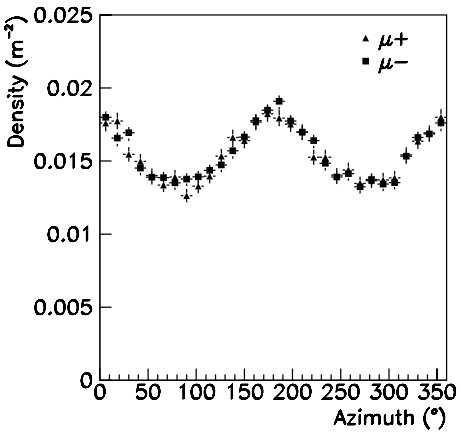


Fig. 8 – The mean azimuthal μ^+- and μ^--distributions of proton induced EAS of inclined showers ($\theta = 45°$) incident from south with the primary energy of 10^{15} eV at a distance of $45 - 50$ m from the shower axis. Earth's magnetic field switched off.

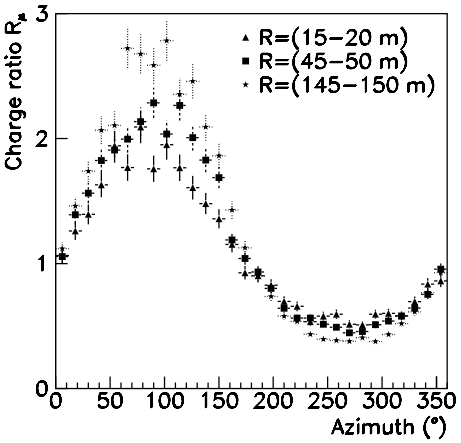


Fig. 9 – Variation of the charge ratio R_μ of the mean muon density distribution of proton induced EAS of inclined showers ($\theta = 45°$) incident from north with the primary energy of 10^{15} eV at various distances R from the shower axis.

- The azimuthal variation of the muon density and of local muon charge ratio in EAS is strongly influenced at the considered energies by the geomagnetic field (compare Figs. 5 and 9).
- This influence depends from the azimuthal direction of the EAS incidence i.e. the direction relative to the direction of the Earth magnetic field and it increased with the distance from the shower axis (compare Figs. 9 and 10).

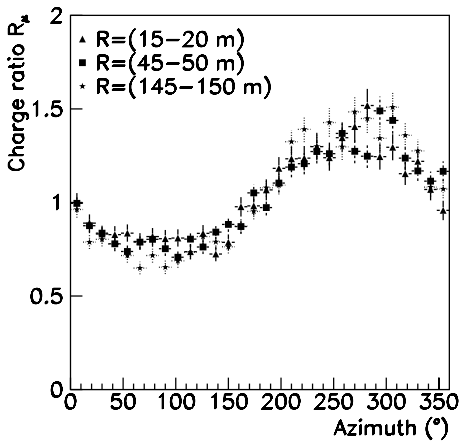


Fig. 10 – Variation of charge ratio R_μ of the mean muon density distribution of proton induced EAS of inclined showers ($\theta = 45°$) incident from south with the primary energy of 10^{15} eV at various distances R from the shower axis.

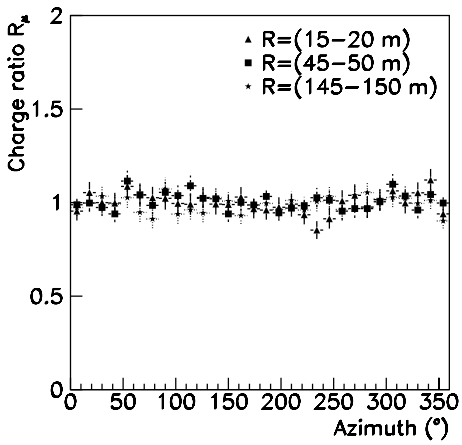


Fig. 11 – Variation of charge ratio R_μ of the mean muon density distribution of proton induced EAS of inclined showers ($\theta = 45°$) incident from north with the primary energy of 10^{15} eV at various distances R from the shower axis. Earth's magnetic field switched off.

- In case of neglect of the geomagnetic field the azimuthal variation in the observation plane reduces to the variation due to geometric effects and for larger radial distances from the core also due to the attenuation effect (see Fig. 4) which does not influence differently positive and negative muons (consequently the value of R^μ remains nearby unity, since obviously differences in the π^+-

and π^--productions remain rather small at the considered muon energies, see Fig. 11).

It should be additionally remarked that for proton induced showers (incident from north or south) the muon charge ratio, integrated over all distances results in $R_\mu = 1.028 \pm 0.002$ (for Fe induced EAS in $R_\mu = 1.025 \pm 0.0015$), where the uncertainty is the standard error for a typical EAS.

5. CONCEPT OF EXPERIMENTAL APPROACHES

Experimental studies of the azimuthal and radial variations of the charge ratio of the muon density in EAS would need a device (spectrometer) for the determination of R_μ placed inside or nearby a detector array for determining the incident EAS, the direction of incidence, defining also the azimuth position (relative to the plane of shower incidence) of the spectrometer and the position of the shower core relative to the location of the spectrometer.

The group of the Okayama University, Japan [37, 38] has installed on the rooftop of a campus building a small array of 8 plastic scintillators, named OU array, to observe extensive air showers. In addition there is a magnetic spectrometer (Okayama telescope) located under the OU array to spectroscopy positive and negative muons, especially to determine the charge sign from the curvature of the tracks of the incident muons and the charge excess of the muons with a momentum larger than 1 GeV/c. The apparatus could be used for coincidence observation of the EAS array with the magnetic spectrometer. First results of the analyses of measured data indicate a charge excess, disappearing within the (large) error bars. But there has been neither the azimuth location of the spectrometer relative to the shower plane nor the angles of the incident EAS specified, and the results may be understood as an average cancelling out the effects. Also the fact that the Okayama telescope finds a rather small value of the muon charge ratio of atmospheric muons may cast some doubts on the systematic accuracy of the measurements.

The spectrometer WILLI [2, 39, 28] installed in Bucharest, Romania determines the muon charge ratio by measuring the life time of the muons stopped in a stack of 16 detector layers. The life time is different for positive and negative muons as negative muons can be captured in atomic orbits, thus leading to an effectively reduced lifetime, depending on the stopping material. The uncertainties of the efficiencies for the detection of differently charged muons and of the geometrical acceptance, usually affecting magnetic spectrometers, are cancelled out. The detector area of the device is able to be directed in a particular direction to measure the incoming muons under different (zenith and azimuth) angles of incidence. Actually WILLI has provided

results of successful measurements, resulting in R_μ-values well in the range of the world average, down to a momentum range of about 0.2 MeV/c. It has demonstrated the azimuthal variation (east-west effect) of the muon charge ratio of atmospheric muons.

There is a proposal [28] to equip the WILLI device with an mini-array of 12 scintillator units, located in distances form about 5 m to about 40 m nearby WILLI, in order to determine the EAS core position relative to the WILLI device and the direction of incidence of showers of the energy region of $10^{14} - 10^{15}$ eV, providing a trigger for WILLI. Fig. 12 displays a sketch of a geometrical layout of the detector system which is under consideration by simulation studies with respect to the efficiency and the detector responses. Though such a small array will not be able to determine the energy of the

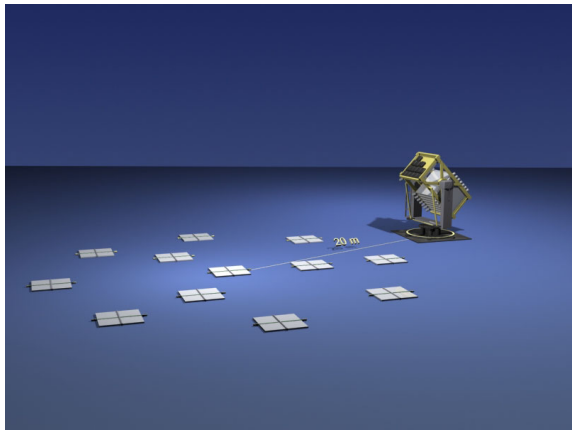


Fig. 12 – Sketch of the geometrical layout of a mini-array for triggering the nearby located WILLI device for muon detection [28].

incident showers, it is expected that such detector system will provide more detailed information about the different propagation of positive and negative EAS muons, in particular under the influence of the geomagnetic field.

6. CONCLUDING REMARKS

EAS simulations show that the lateral density distribution of positive and negative muons are varying not only with the (radial) distance from the shower axis, but also with the azimuth relative to the plane of the incident shower. The reasons are different. In addition to the attenuation effects of charged particles of inclined showers [15] in the atmosphere by the variation of the travelling distances in the atmosphere, the geomagnetic field affects the travel

of positive and negative muons in an opposite way. The geomagnetic effects depend on the direction of the EAS axis relative to the Earth's magnetic vector. This leads to an azimuthal variation of the muon charge ratio of the muon density distribution. In the extreme case of very inclined showers (with long slant depths) the Earth's field could be used as magnetic separator [29, 30]. Obviously the experimental detection of these features is of great interest for the understanding of the EAS development. Furthermore the quantitative results would also provide some information about the hadronic interaction, in particular when observing higher energy muons [20]. The aspect of the dependence on the hadronic interaction models, currently en vogue [40], has not been systematically explored in the present paper.

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